



VIT-AP
UNIVERSITY

Module 2: Cloud-Enabling Technologies

Virtualization Technology

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OUTLINE

- **Cloud-Enabling Technologies**

Broadband Networks and Internet Architecture

foundational network and internet technologies that enable cloud computing, including the role of broadband in providing high-speed data access.

Data Center Technology

Covers the technology and infrastructure used in data centers, which are critical to cloud computing as they house the servers, storage, and networking equipment that deliver cloud services.

Virtualization Technology

that allows multiple virtual machines (VMs) to run on a single physical machine, enabling efficient resource utilization and scalability in cloud environments.

Web Technology

Focuses on the web-based technologies that support cloud computing, such as **web servers, APIs, and protocols** that facilitate interaction between cloud services and users.

Multitenant Technology

Describes the technology that allows multiple users or tenants to share the same computing resources in a cloud environment while keeping their data and applications isolated from each other.

Service Technology

- **Summary**

- **Web Technology**

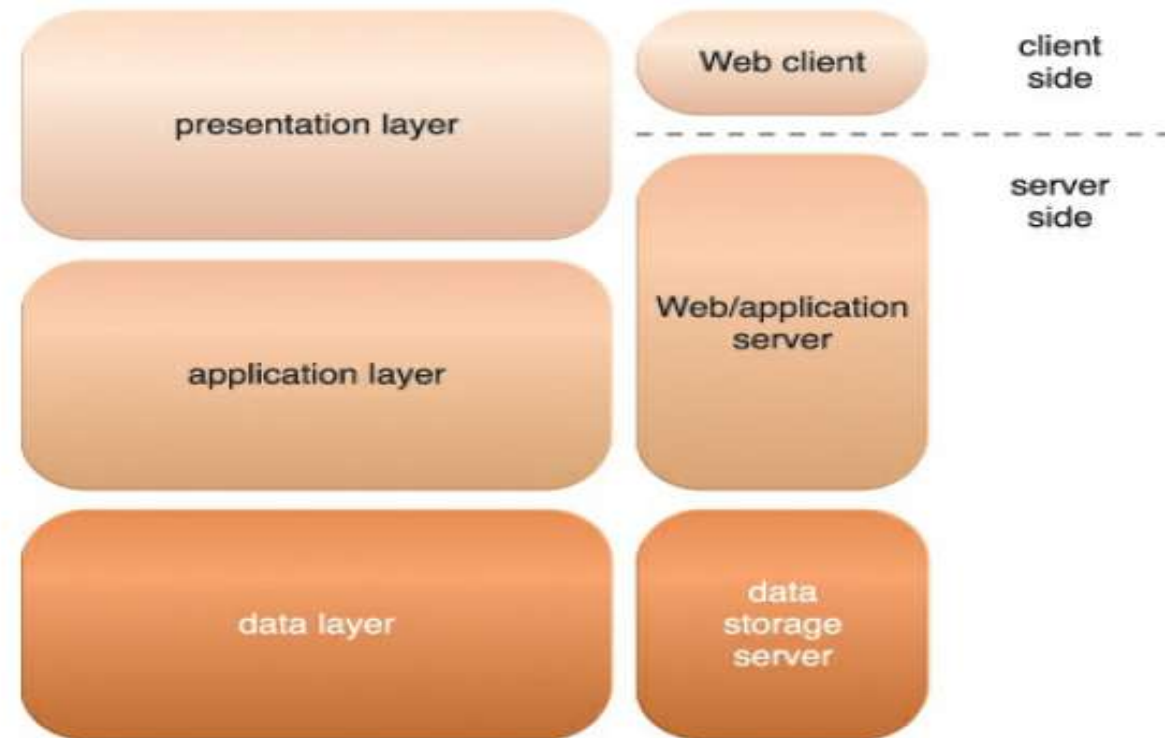
- WWW is a system of interlinked IT resources that are accessed through the Internet
- Basic Components: Web Browser Client, Web Server
- Additional Components (for scalability and security):
 - Proxies, Caching Services, Gateways, Load Balancers
- **Technology Architecture of the Web:**
 - Uniform Resource Locator (URL): Standard syntax for creating identifiers.
 - Hypertext Transfer Protocol (HTTP): Main communication protocol for exchanging content and data on the Web; URLs are transmitted via HTTP.
 - Markup Languages (HTML, XML):
 - HTML: Expresses Web page presentation.
 - XML: Defines vocabularies to associate meaning with **Web data via metadata.**

- **Example Process:**

- Web browser requests an action (e.g., read, write) on a Web resource via its URL.
- Request sent via HTTP to the resource host.
- Web server locates the resource, performs the action, and responds to the client.
- Response may include HTML and XML content.

- **Web Application:**

- A distributed application using Web-based technologies, typically relying on Web browsers for user interfaces.
- Commonly **found** in cloud-based environments due to high accessibility.

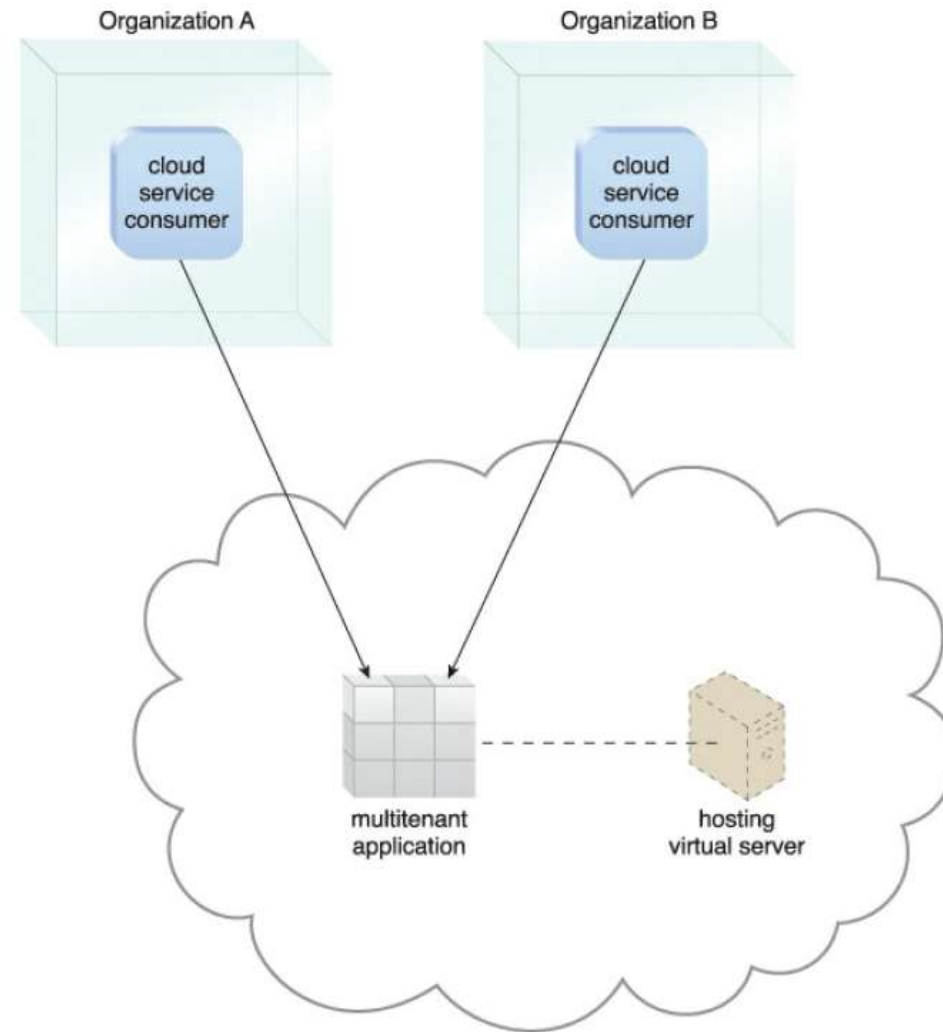


- **Architecture**
 - **Presentation Layer:**
 - User interface; includes components on both client and server sides.
 - Receives client requests.
 - **Application Layer:**
 - Implements application logic.
 - Retrieves resources as static Web content or generates dynamic Web content based on application logic.
 - **Data Layer:**
 - Comprises persistent data stores.
 - Interacts with application servers to execute application logic, often involving database interaction.
 - **PaaS (Platform as a Service):**
 - Provides ready-made environments for developing and deploying Web applications.

- ❑ Web Technology
- ❑ Web Application:
- ❑ Multitenant Technology

Multitenant Technology

- Multitenant Technology



Multitenant Technology

- ☐ Web Technology
- ☐ Web Application:
- ☐ Multitenant Technology

- **Multitenant Technology**
 - Allows multiple users (tenants) to access the same application logic simultaneously.
 - Each tenant views, administers, and customizes the application as a dedicated instance, **unaware** of other tenants.
 - Tenants cannot access **data or configurations** of other tenants.
 - Customizable **Features** for Tenants:
 - **User Interface:** Customize the look and feel.
 - **Business Process:** Modify rules, logic, and workflows.
 - **Data Model:** Extend or alter the data schema (e.g., include, exclude, rename fields).
 - **Access Control:** Manage user and group access rights.

Multitenant Technology

- ❑ Web Technology
- ❑ Web Application:
- ❑ Multitenant Technology

- **Characteristics**

- **Usage Isolation:** One tenant's behavior does not impact the availability or performance for others.
- **Data Security:** Tenants are restricted from accessing each other's data.
- **Recovery:** Backup and restore procedures are tenant-specific.
- **Application Upgrades:** Upgrades to shared software do not negatively affect tenants.
- **Scalability:** The application can scale for increased usage or additional tenants.
- **Metered Usage:** Tenants are billed based on actual usage of processing and features.
- **Data Tier Isolation:** Tenants can have isolated or shared databases, tables, and schemas.

- ☐ Web Technology
- ☐ Web Application:
- ☐ Multitenant Technology
- ☐ Service Technology

- Service Technology

- Refers to the various methods and technologies used to create, manage, and deploy services in a networked environment.
- Web services are standardized ways of integrating web-based applications using XML, SOAP, WSDL, and UDDI open standards over an internet protocol backbone.
 - SOAP (Simple Object Access Protocol): Protocol for exchanging structured information in web services.
 - WSDL (Web Services Description Language): XML-based language for describing web services.
 - UDDI (Universal Description, Discovery, and Integration): A directory service where businesses can register and find web services.
 - REST Services Definition: Representational State Transfer (REST) is an architectural style for building web services that use HTTP requests for communication.

Service Technology

- ❑ Web Technology
- ❑ Web Application:
- ❑ Multitenant Technology
- ❑ Service Technology

- Service Agents
 - Service agents are intermediary software components that act on behalf of service consumers or providers to perform tasks such as monitoring, filtering, logging, or adapting services.
 - Functions:
 - Monitoring: Track and report the performance and availability of services.
 - Adapting: Transform requests and responses to ensure compatibility between service consumers and providers.
 - Security Enforcement: Ensure that only authorized requests reach the service.
- Service Middleware
 - Definition: Service middleware is software that lies between the client and the server, facilitating communication, integration, and management of services.
 - Roles:
 - Communication Handling: Manages data exchange and message routing between services.
 - Service Integration: Connects disparate systems and services, enabling them to work together.
 - Transaction Management: Ensures that all parts of a service-based transaction are completed successfully.
 - Security and Authentication: Provides mechanisms for securing service interactions and authenticating users.

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- ☐ Web Application:
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- ☐ **Virtualization Technology**
- ☐ **Key points**
- ☐ **Hardware Independence**
- ☐ **Server Consolidation**
- ☐ **Resource Replication**
- ☐ **Operating System-Based Virtualization**
- ☐ **Hardware-Based Virtualization**
- ☐ **Virtualization Management**
- ☐ **Other Considerations**

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Quiz

**Thank
You**